

Prefix Sum - Complete Notes

What is Prefix Sum?

Prefix Sum stores cumulative sums of an array. It helps answer range-sum queries efficiently.

$\text{prefix}[i]$ = sum of elements from index 0 to i

When to Use Prefix Sum

Use it for Range Sum Queries, $\text{Sum}(L,R)$, Multiple Queries, and Subarray Sum calculations.

Syntax

```
prefix[0] = arr[0];
for(int i=1;i<arr.length;i++){
    prefix[i]=prefix[i-1]+arr[i];
}
```

Basic Code

```
int[] arr = {2,4,1,7,3,6};
int[] prefix = new int[arr.length];
prefix[0] = arr[0];
for(int i=1;i<arr.length;i++)
{
    prefix[i] = prefix[i-1] + arr[i];
}
```

Dry Run

Array: [2,4,1,7,3,6]

Prefix: [2,6,7,14,17,23]

Range Sum Formula

$\text{sum}(L,R) = \text{prefix}[R] - \text{prefix}[L-1]$

If $L == 0 \rightarrow \text{sum} = \text{prefix}[R]$

Example

$\text{sum}(1,4) = \text{prefix}[4] - \text{prefix}[0] = 17 - 2 = 15$

Complexity

Build: $O(n)$

Query: $O(1)$

Space: $O(n)$

Common LeetCode Questions

303. Range Sum Query - Immutable

560. Subarray Sum Equals K

724. Find Pivot Index

Recognition Trick

Range Sum? L to R? Multiple Queries? -> PREFIX SUM